

$$\rightarrow \langle N, S, u \rangle$$

6. Static games of incomplete information

- So far, we assumed that players have complete information about the game so that, in particular, every player knows
 - ▶ every other player's preferences (incl. rationality)
 - ▶ every other player's strategy space
 - ▶ every other player's information partition
- In many contexts, this asks for too much:
 - ▶ A potential market entrant may not know if the incumbent would prefer *accommodate* or *fight* after entry
 - ▶ A seller typically doesn't know a potential buyer's willingness to pay when both bargain over price
- What use are NE or SPE, which rest on correct beliefs about others' behavior in the game, when there is *incomplete information* even about which game is being played?

6.1 Harsanyi's transformation

- John C. Harsanyi (1967/68) proposed a powerful framework for games of incomplete information
 1. Introduce different *types* of each player:
 - ▶ A particular type θ_i of player i is identified with a particular preference, strategy space, and information partition
 - ▶ Each player i knows his own type θ_i but possibly not that of other players
 2. Introduce an additional player, called *Nature*:
 - ▶ *Nature* moves before everybody else and assigns each player i his type $\theta_i \in \Theta_i$
 - ▶ Nature's move is a random draw from an exogenous joint probability distribution ρ on $\Theta \equiv \Theta_1 \times \dots \times \Theta_n$ assumed to be commonly known
 - ▶ Each player i rationally updates the common prior ρ after learning θ_i
- Thus, a game of *incomplete information* is transformed into an (extensive) game with *complete (but imperfect) information*

Example I

- Suppose a potential entrant and the incumbent simultaneously decide about whether to enter and whether to boost capacity, respectively
- Cost of a capacity increase is either high or low, and *private information* of (=known only to) the incumbent
- Profits are

Incumbent \ Entrant	enter	stay out
invest	0,-1	2,0
don't invest	2,1	3,0

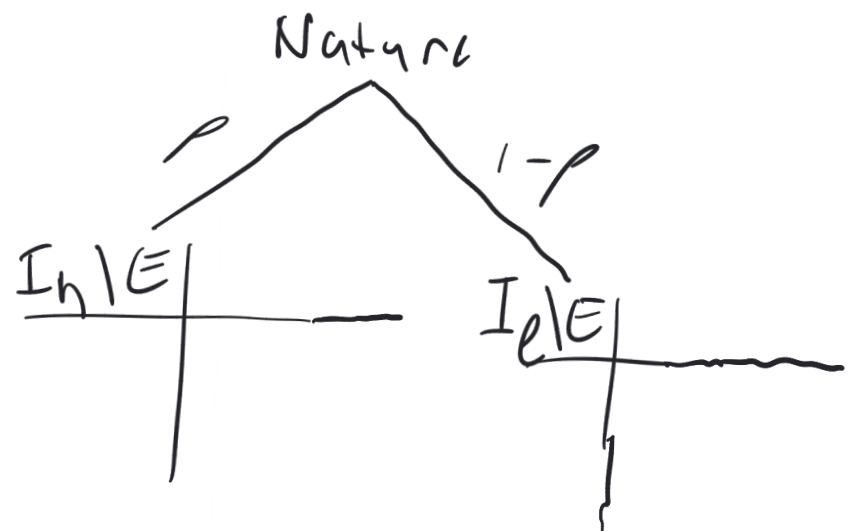
in case of *high costs*

and

Incumbent \ Entrant	enter	stay out
invest	3,-1	5,0
don't invest	2,1	3,0

in case of *low costs*

$(D, I; \text{enter})$ if $p \geq \frac{1}{2}$
 $(D, I; \text{stay out})$ if $p \leq \frac{1}{2}$



$$u_E(\text{enter}) = p \cdot 1 + (1-p) \cdot (-1)$$

Analysis of Example I

- Firm 1 has a strictly dominant strategy:
 - ▶ *Invest* when cost is low
 - ▶ *Don't invest* when cost is high
- However, firm 2 does not know whether firm 1 has high or low costs
- Firm 2 assigns a *prior probability* to firm 1's cost being high: Let $\rho \in (0, 1)$ be the probability that firm 2 assigns to firm 1 having a high cost of building a new plant (and assume this is common knowledge)
- Firm 2's expected payoff is
 - ▶ $p - (1 - p) = 2p - 1$ if it enters,
 - ▶ and 0 if it stays out
- Therefore, firm 2 will enter the market if and only if

$$2\rho - 1 \geq 0$$

or iff its prior belief about firm 1's cost being high is greater than $\frac{1}{2}$

Example II

- Consider the situation from Example I again, but suppose that profits are

→

Incumbent \ Entrant	enter	stay out
invest	0, -1	2, 0
don't invest	2, 1	3, 0

in case of *high costs*

and

$\langle q \rangle$ $\langle 1-q \rangle$

Incumbent \ Entrant	enter	stay out
invest	1.5, -1	3.5, 0
don't invest	2, 1	3, 0

in case of *low costs*

no strictly dominant strategy

1) I_n will always choose D

2) Let q denote the prob. the incumbent allows to "entry"

$$u_{II}(\text{invest})$$

$$= q \cdot 1.5 + (1-q) \cdot 3.5$$

$$u_{II}(\text{don't invest}) = q \cdot 2 + (1-q) \cdot 3$$

$$BR_I(q, 1-q) = P(q) = \begin{cases} \text{invest} & \text{if } q < \frac{1}{2} \\ [0, 1] & \text{if } q = \frac{1}{2} \\ \text{don't invest} & \text{if } q > \frac{1}{2} \end{cases}$$

Example II - continued

- Consider the situation from Example I again, but suppose that profits are

Incumbent \ Entrant	enter	stay out
invest	0, -1	2, 0
don't invest	2, 1	3, 0

in case of high costs

and

Incumbent \ Entrant	enter	stay out
invest	1.5, -1	3.5, 0
don't invest	2, 1	3, 0

in case of low costs

no strictly dominant strategy

Consider $\frac{1}{2(1-p)}$

$$\text{if } p = \frac{1}{2} \Rightarrow \frac{1}{2(1-p)} = 1$$

$$p > \frac{1}{2} \Rightarrow \text{entrant enters for sure}$$

$$p < \frac{1}{2} \Rightarrow \frac{1}{2(1-p)} < 1$$

$$u_E(\text{enter} | p) \geq u_E(\text{stay out} | p) \quad \text{stay out} \downarrow$$

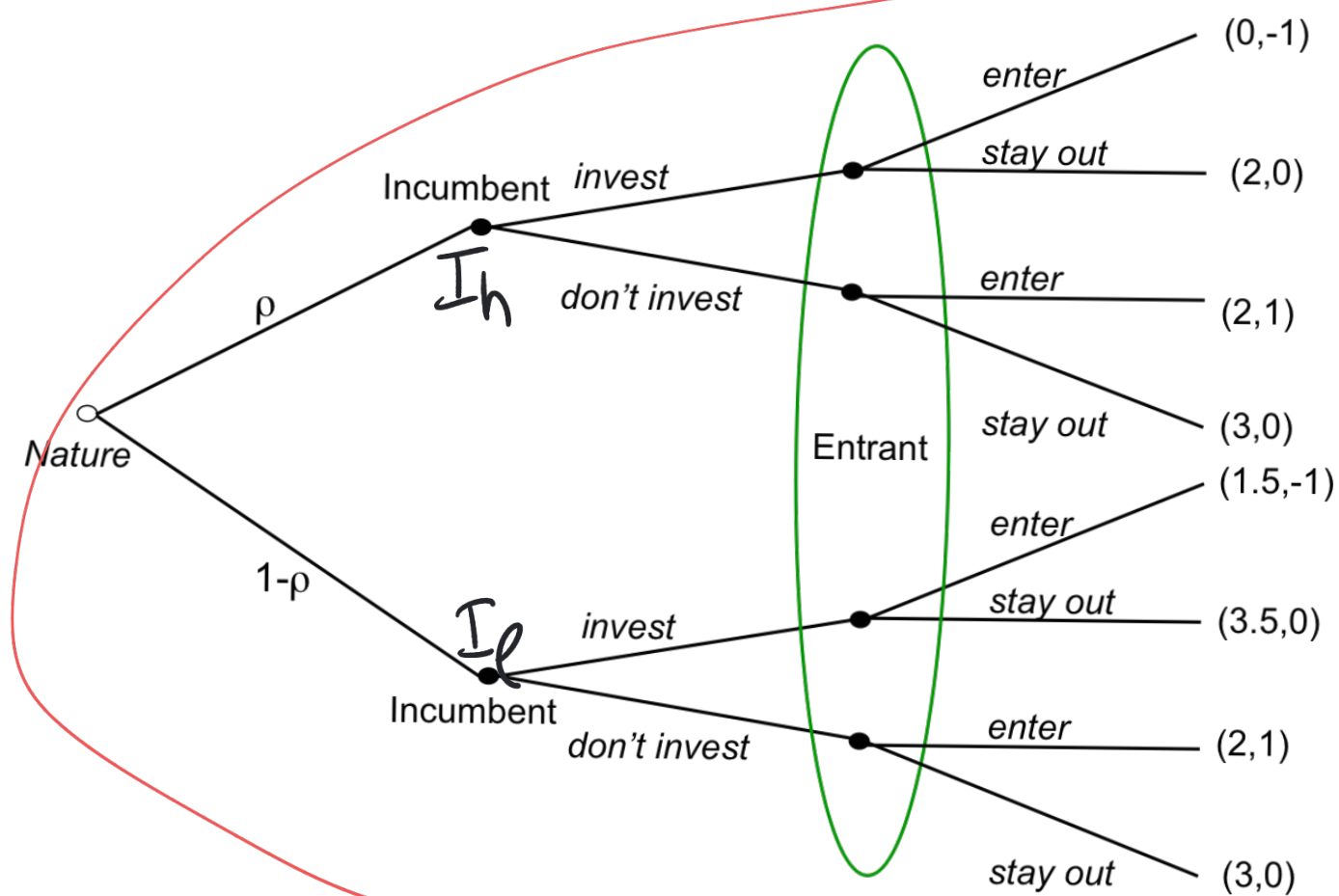
$$p \cdot 1 + (1-p)[p(-1) + (1-p)1] \geq 0$$

$$\hookrightarrow \text{enter if } p \leq \frac{1}{2(1-p)}$$

$$BR_E(p, 1-p) = \begin{cases} \text{enter} & \text{if } p < \frac{1}{2(1-p)} \\ [0, 1] & \text{if } p = \frac{1}{2(1-p)} \\ \text{stay out} & \text{if } p > \frac{1}{2(1-p)} \end{cases}$$

Example

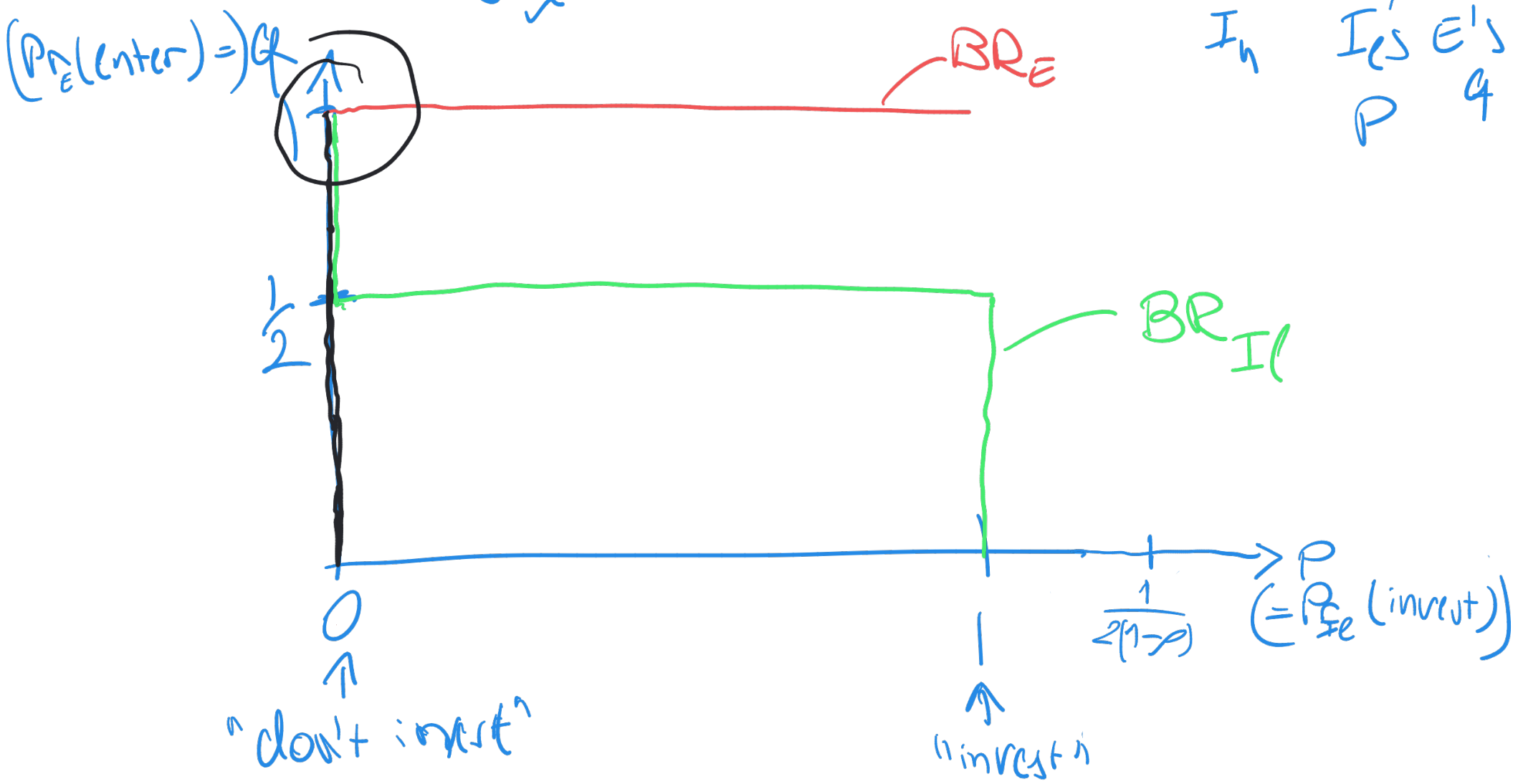
- Assuming that *Nature* 'selects' high costs with probability ρ we obtain the corresponding game of complete information:

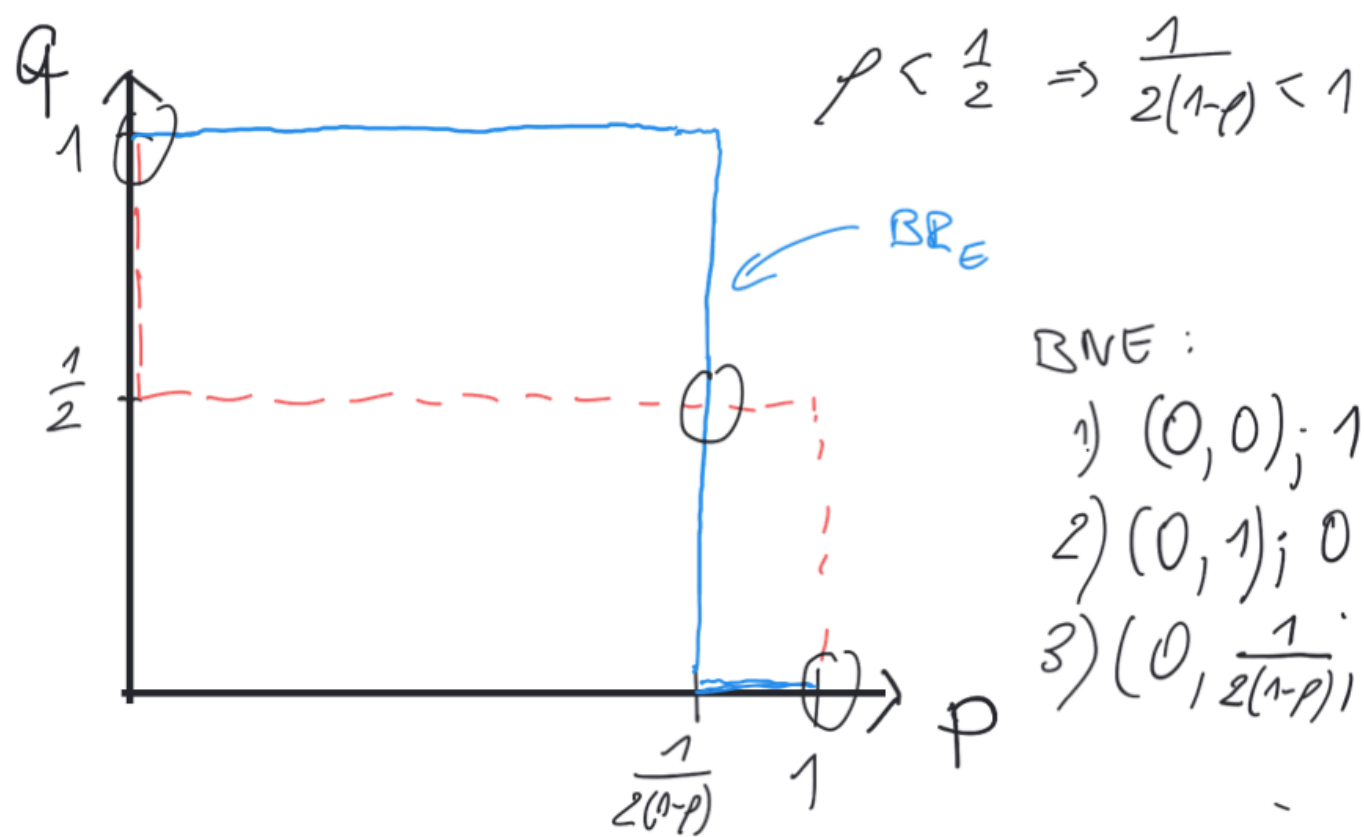


whole game tree is the only subgame

	enter	stay out
I_h		
I_l		
D		
I		
D		

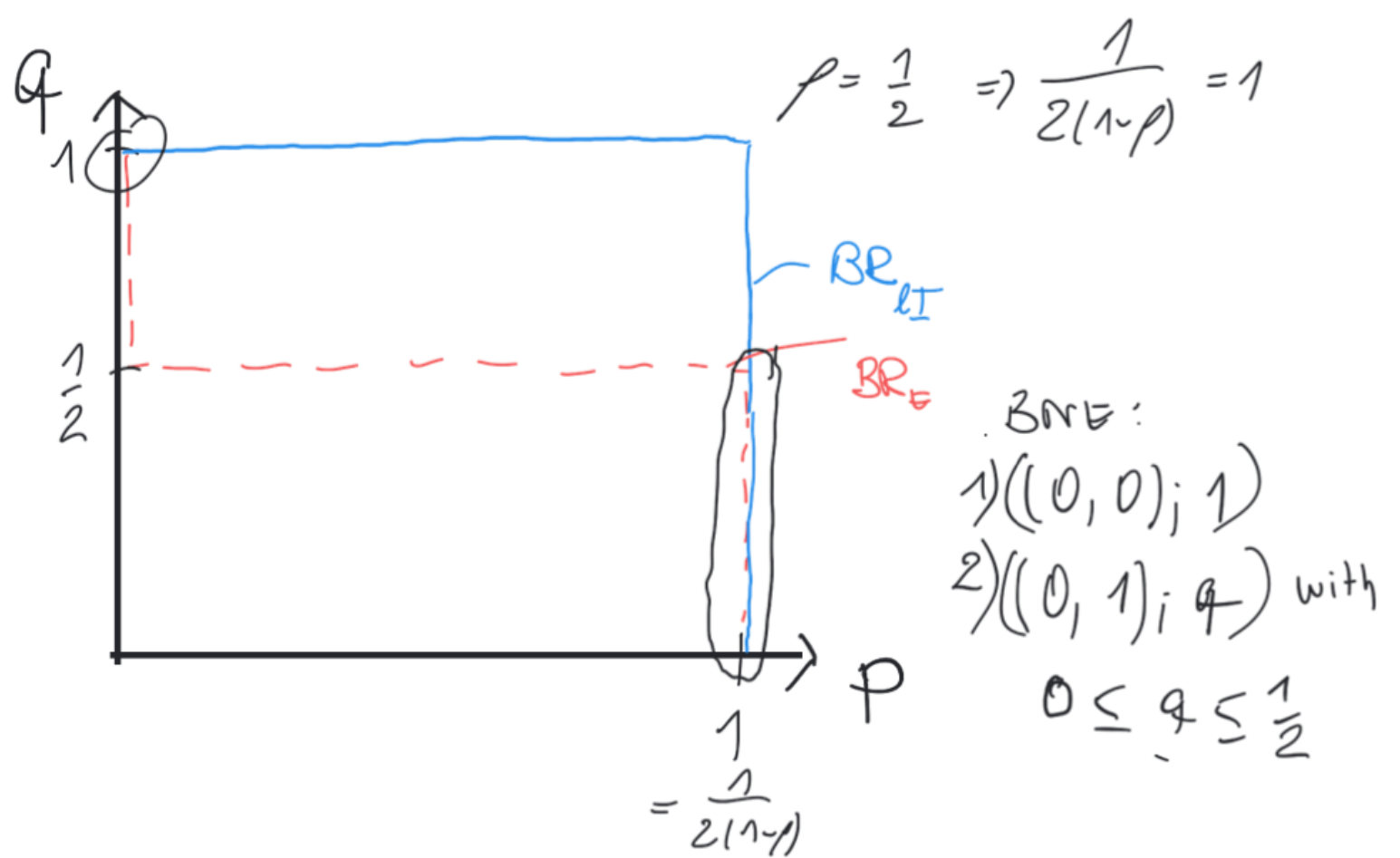
$\rho > \frac{1}{2}$
 $\underset{\substack{\uparrow \\ \text{Bayesian}}}{\text{BNE}} (DD; \text{enter}) = (0, 0, 1)$
 $= (\underset{\substack{\uparrow \\ I_h}}{0}, \underset{\substack{\uparrow \\ I_h' \cdot \frac{1}{P}}}{0}, \underset{\substack{\uparrow \\ E_h' \cdot \frac{1}{4}}}{1})$





BNE :

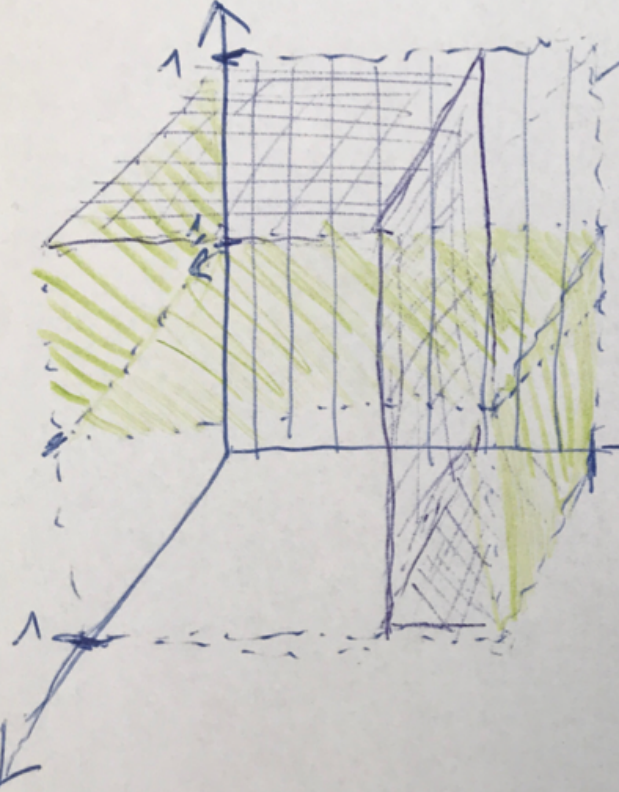
- 1) $(0, 0); 1$
- 2) $(0, 1); 0$
- 3) $(0, \frac{1}{2(1-p)}, \frac{1}{2})$



$q = \text{Prob}(\text{entrant chooses "enter"})$

$P_A = 0$: don't invest
is strictly dominant

Case: $p < \frac{1}{2}$



$P_L = \text{Prob}(\text{low cost incumbent chooses "invest"})$

$P_H = \text{Prob}(\text{high cost incumbent chooses "invest"})$

Best-response diagram
for incumbent/entrant game

6.2 Bayesian games

$\langle N, S, u \rangle$

Definition 6.1

A *Bayesian (normal form) game* is a collection $\langle \overset{\checkmark}{N}, \overset{\checkmark}{\Theta}, \overset{\checkmark}{\rho}, \overset{\checkmark}{\mathbf{A}}, \overset{\checkmark}{\mathbf{u}} \rangle$ where

- type space $\Theta \equiv \Theta_1 \times \dots \times \Theta_n$ specifies all possible types of players $i \in N$
- actual types are drawn from the joint probability distribution ρ on Θ
- players' (pure) strategy spaces S_i are implicitly defined as the set of all functions $s_i : \Theta_i \rightarrow A_i$ which map every possible type of player i , θ_i , to an action $s_i(\theta_i) \in A_i$
(elements of A_i are strategies in the original game of incomplete information)
- u_i is defined on $\mathbf{A} \times \Theta_i$

Common prior

- We assume that $\langle N, \Theta, \rho, \mathbf{A}, \mathbf{u} \rangle$ incl. players' rationality is common knowledge
- We assume that all players have the same beliefs about how types are distributed (the *common prior* ρ)
- Given $\rho(\theta_1, \dots, \theta_n)$ we can compute the conditional distribution

$$\rho(\theta_{-i}|\theta_i) = \frac{\rho(\theta_{-i}, \theta_i)}{\rho(\theta_i)}$$

using Bayes' Rule

- Many applications assume that players' types are *independent*, i.e.

$$\rho(\theta_{-i}|\theta_i) = \frac{\rho(\theta_{-i}) \rho(\theta_i)}{\rho(\theta_i)} = \rho(\theta_{-i})$$

(does not depend on θ_i)

- Players update the **common prior** ρ using *Bayes' rule*:

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}$$

Remarks

- The assumption of a correct common prior ρ about all players' types is *not at all* innocuous
(somewhat more generally, one can work with so-called *global types* θ_i which also contain a full description of player i 's beliefs about all players' types, his beliefs about their beliefs about types, his beliefs about their beliefs about his beliefs about types, etc.)
- A player's preference over (post-*Nature*) terminal histories usually is a function of his own type, but not of other players' types
(i 's payoff is affected by their types only indirectly via their play)

Extra remarks

- In the mixed extension of $\langle N, \Theta, \rho, \mathbf{A}, \mathbf{u} \rangle$ players can independently randomize over A_i , i.e., mixed strategies are functions $\sigma_i : \Theta_i \rightarrow \Delta(A_i)$
- Equivalently to including common prior ρ in the game definition, one may specify a profile $\boldsymbol{\rho} = (\rho_1, \dots, \rho_n)$ of functions $\rho_i : \Theta_i \rightarrow \Delta(\Theta_{-i})$ which describe players' beliefs conditional on the own type (derived from ρ by Bayes' rule);
for independent types, ρ_i is the same for any θ_i
- Different action sets for different types of the same player can be accounted for by including *all* actions in A_i but having $u_i(a_i, a_{-i}; \theta_i) = -\infty$ if a_i is unavailable to θ_i

Example

- Consider $\langle N, \Theta, \rho, \mathbf{A}, \mathbf{u} \rangle$ with
 - ▶ $N = \{1, 2\}$
 - ▶ $\Theta = \{1_h, 1_l\} \times \{2\}$
 - ▶ ρ with $\rho(1_h, 2) = \rho(1_l, 2) = 0.5$
 - ▶ $\mathbf{A} = \{\text{invest, don't invest}\} \times \{\text{enter, stay out}\}$
 - ▶ $\mathbf{u}$ as specified by the table

$1_h/1_l \ \backslash \ 2$	<i>enter</i>	<i>stay out</i>
<i>invest</i>	0/1.5,-1	2/3.5,0
<i>don't invest</i>	2/2,1	3/3,0

6.3 Bayesian Nash equilibrium

- Strategies specify what each player would do, in any possible contingency in which he might be called upon to play
 - ▶ A player's strategy has to do so for each possible type of that player:
 - ▶ Think of each type of each player as a separate incarnation of that player
 - ▶ Nature makes the first move and, using probability distribution ρ , decides randomly which of these incarnations will actually play the game

More formally

- Comparing two actions $a_i, a'_i \in A_i$, player i with type θ_i will (in equilibrium: correctly) anticipate some *strategy* profile s_{-i} but – in the spirit of players having incomplete information – must treat other players' *types* and hence actions as random variables, i.e., type θ_i compares

$$\mathbf{E}u_i(a_i, s_{-i}; \theta_i) \equiv \sum_{\theta_{-i} \in \Theta_{-i}} \rho(\theta_{-i} | \theta_i) \cdot u_i(a_i, s_{-i}(\theta_{-i}); \theta_i)$$

to $\mathbf{E}u_i(a'_i, s_{-i}; \theta_i)$

- Strategy s_i^* of player i (in a Bayesian game) is a *best response* to s_{-i} iff it specifies an optimal action $s_i^*(\theta_i) \in A_i$ for *any* type θ_i that player i might happen to be,
i.e., $\forall \theta_i \in \Theta_i : \forall a'_i \in A_i : \mathbf{E}u_i(s_i^*(\theta_i), s_{-i}; \theta_i) \geq \mathbf{E}u_i(a'_i, s_{-i}; \theta_i)$

Bayesian Nash equilibrium

- The concept of a strategy profile in which each player best-responds to the other players (= NE) is now straightforwardly extended to Bayesian games:

Definition 6.2

A *Bayesian Nash equilibrium (BNE)* of the game $\langle N, \Theta, \rho, \mathbf{A}, \mathbf{u} \rangle$ is a strategy profile $\mathbf{s}^* = (s_1^*, \dots, s_n^*)$ such that for any player $i \in N$ the strategy s_i^* is a best response to s_{-i}^* , i.e.,

$$\forall \theta_i \in \Theta_i : s_i^*(\theta_i) \in A_i \text{ maximizes } \mathbf{E} u_i(a_i, s_{-i}^*; \theta_i)$$

(with expectation $\mathbf{E}$ based on $\rho(\theta_{-i}|\theta_i)$)

Remarks

- A mixed-strategy BNE σ^* is defined analogously
- As in games of complete information, mixed strategy σ_i^* is a best response to σ_{-i} iff each action a_i played with probability $\sigma_i(\theta_i)(a_i) > 0$ maximizes $\mathbf{E}u_i(a_i, \sigma_{-i}; \theta_i)$
- Proofs of existence of BNE are analogous to those for NE (player i 's best response correspondence BR_i is now $BR_i(\theta_1) \times BR_i(\theta_2) \times \dots \times BR_i(\theta_{k_i})$)

Example II again

$1_h/1_l \setminus 2$	<i>enter</i>	<i>stay out</i>
<i>invest</i>	0/1.5,-1	2/3.5,0
<i>don't invest</i>	2/2,1	3/3,0

with probability $\rho \in [0, 1]$ for firm 1 having high costs

- What are the BNE ...

- ▶ if $\rho > 0.5$?
- ▶ if $\rho < 0.5$?
- ▶ if $\rho = 0.5$?

0⁺ type

37

 $\frac{1}{67}$

2,0

 $\frac{1}{3} \times$ $\frac{1}{2}$ ~~456~~0,0

- Handwritten notes on a grid:
- Top-left: $\odot / \odot$
 - Top-right: FT
 - Middle-left: $F\bar{F}$
 - Bottom-left: TF
 - Bottom-right: (10)

Example III – analysis

	FF	FT	TF	TT
FF	①	②	③	④
FT	⑤	⑥	⑦	⑧
TF	⑨	⑩	⑪	⑫
TT	⑬	⑭	⑮	⑯

$$\textcircled{1} \quad u_f(FF; FF) = 4 \cdot \frac{1}{3} + 4 \cdot \frac{1}{3} + 0 \cdot \frac{1}{6} + 0 \cdot \frac{1}{6} = \frac{8}{3}$$

$$\textcircled{10} \quad u_f(TF; FT) = 0 \cdot \frac{1}{3} + 2 \cdot \frac{1}{3} + 0 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} = \frac{4}{3}$$

pure BNE: 1) (FF; FT)
2) (TF; TT)

	FF	FT	TF	TT
FF	8/3, 1	2, 3	2, 0	2, 2
FT	10/3, 1	5/3, 2	5/3, 1	0, 2
TF	0, 1	4/3, 1	4/3, 2	8/3, 2
TT	2/3, 1	1, 0	1, 3	4/3, 2

6.4 Purification of mixed strategies (due to Harsanyi)

Idea:

- One can interpret a mixed strategy eq'm in a game with complete information as an eq'm in pure strategies in a game with incomplete information
- That is, the crucial feature of a mixed-strategy NE is not that player j chooses a strategy randomly, but rather that player i is uncertain about player j 's choice
- This uncertainty arises because of slight incomplete information regarding others' preferences

Consider the game "Battle of the sexes" but assume now that *player 2 does not know the payoffs of his opponent exactly*:

1\2		Opera	Fight				
Opera		$2 + t_1, 1$	$0, 0$	$\begin{pmatrix} 2 \\ 3 \end{pmatrix} O$	$2, 1$	$0, 0$	$\Delta 1$
Fight		$0, 0$	$1, 2 + t_2$	$\begin{pmatrix} 4 \\ 3 \end{pmatrix} F$	$0, 0$	$1, 2$	$\Delta 2$

Analysis of ‘perturbed’ BoS

- What are the players’
 - ▶ Action spaces: $A_i \in \{O, F\}$
 - ▶ Type spaces? $t_i \sim U[0, x]$
 - ▶ Prior probability distributions? $f(t_i) = \frac{1}{x}$ if $t_i \in [0, x]$ and 0 otherwise
 - ▶ Payoffs? See table
- Construct an eq’m in pure cutoff strategies in which
 - ▶ σ_1 : Player 1 chooses Opera if $t_1 \geq c_1$ and Fight otherwise
 - ▶ σ_2 : Player 2 chooses Fight if $t_2 \geq c_2$ and Opera otherwisewhere $c_1 > 0$ and $c_2 > 0$ are some cutoff values that we will determine

Analysis of ‘perturbed’ BoS

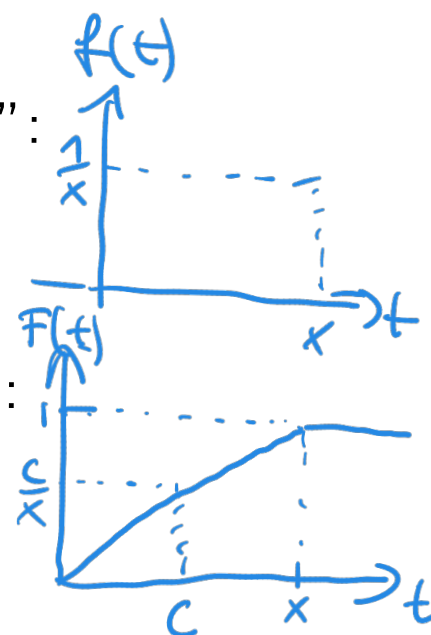
- In this equilibrium, the probability

- ▶ Player 2 attaches to the event “player 1 chooses Opera”:

$$Prob(t_1 \geq c_1) = 1 - F(c_1) = 1 - \frac{c_1}{x}$$

- ▶ Player 1 attaches to the event “player 2 chooses Fight”:

$$Prob(t_2 \geq c_2) = 1 - F(c_2) = 1 - \frac{c_2}{x}$$



- Player 1 prefers Opera over Fight iff: $\bar{u}_1(O, \sigma_2, t_1) \geq \bar{u}_1(F, \sigma_2, t_1)$

$$\Leftrightarrow \underbrace{\frac{c_2}{x}}_{Prob(2 \text{ chooses } O)} (2 + t_1) + \underbrace{\left(1 - \frac{c_2}{x}\right)}_{Prob(2 \text{ chooses } F)} \cdot 0 \geq \frac{c_2}{x} \cdot 0 + \left(1 - \frac{c_2}{x}\right) \cdot 1$$

$$\Leftrightarrow t_1 \geq \frac{x}{c_2} - 3$$

$\Rightarrow c_1 = \frac{x}{c_2} - 3$ for strategy σ_1 of player 1 to be a best response to σ_2

Analysis of ‘perturbed’ BoS

- Player 2 prefers Fight over Opera iff: $\bar{u}_2(O, \sigma_1, t_2) \leq \bar{u}_2(F, \sigma_1, t_2)$

$$\Leftrightarrow \underbrace{\left(1 - \frac{c_1}{x}\right)}_{\text{Prob}(1 \text{ chooses } O)} \cdot 1 + 0 \geq \underbrace{\frac{c_1}{x}}_{\text{Prob}(1 \text{ chooses } F)} (2 + t_2) \Leftrightarrow t_2 \geq \frac{x}{c_1} - 3$$

$\Rightarrow c_2 = \frac{x}{c_1} - 3$ for strategy σ_2 of player 2 to be a best response to σ_1

- Hence, $c_1 = c_2 = c = \frac{x}{c} - 3$

[This follows because our results imply both $c_1 c_2 = x - 3c_1$ and $c_1 c_2 = x - 3c_2 \Rightarrow c_1 = c_2$.]

- $c = \frac{x}{c} - 3 \Leftrightarrow c^2 + 3c - x = 0$
- Solutions to this quadratic equation are:
 - ▶ $c = \frac{\sqrt{9+4x}-3}{2} > 0$, and
 - ▶ $c = \frac{-\sqrt{9+4x}-3}{2} < 0$: not possible as types from $[0, x]$

Analysis of ‘perturbed’ BoS

- What is the probability that player 1 chooses opera?

$$Prob(t_1 \geq c) = 1 - \frac{c}{x} = 1 - \frac{\sqrt{9 + 4x} - 3}{2x}$$

- What happens to this probability if $x \rightarrow 0$?¹

$$\begin{aligned} \lim_{x \rightarrow 0} \left(1 - \frac{\sqrt{9 + 4x} - 3}{2x} \right) &= 1 - \lim_{x \rightarrow 0} \frac{\sqrt{9 + 4x} - 3}{2x} \cdot \overbrace{\frac{\sqrt{9 + 4x} + 3}{\sqrt{9 + 4x} + 3}}^{=1} \\ &= 1 - \lim_{x \rightarrow 0} \frac{9 + 4x - 9}{2x(\sqrt{9 + 4x} + 3)} \\ &= 1 - \lim_{x \rightarrow 0} \frac{2}{\sqrt{9 + 4x} + 3} = 1 - \frac{2}{6} = \frac{2}{3} \end{aligned}$$

- As the incomplete information disappears, behavior in pure-strategy BNE of the incomplete information game approaches behavior in mixed strategy NE in the original game of complete information

¹Derivation notes: $(a - b)(a + b) = a^2 - b^2$, $\lim_{x \rightarrow 0} \sqrt{9 + 4x} = \sqrt{9} = 3$.

6.5 Auctions

- Auctions are a popular means to conduct a transaction with a subset of multiple potential partners
 - ▶ within a reasonably short, often determinate time frame
 - ▶ in a transparent way, which reveals information about the opportunity cost of choosing any particular transaction partner
- Formally, auctions are *anonymous universal mechanisms* that
 - ▶ select an allocation based on information about agents which is elicited (only) via *bids* without special role for bidder's identities
 - ▶ can be used for any type of goods (or 'bads')

Auctions

- Auctioning has a very long history:
 - ▶ Babylonians auctioned wives: the most attractive women were sold first (they commanded positive prices), whereas men were paid to be matched with the least desirable women.
 - ▶ Greeks auctioned mine concessions, rights to collect taxes, to dispose of confiscated property
 - ▶ Romans auctioned slaves
- Modern examples:
 - ▶ Art auctions at Sotheby's or Christie's
 - ▶ Internet auction sites like eBay
 - ▶ Google Ad auction
 - ▶ Spectrum / license / emission auctions
 - ▶ Procurement by tender
(university computers, car parts, ...)

Common types of auctions

- Common types of auctions for *selling* a single indivisible object (with straightforward analogues for *buying* an object):
 - ▶ *English auction* (= open ascending-price auction):
price is publicly raised until all but one player stop expressing interest;
last active bidder receives the object and pays the stop price
[→ see Monet painting auctioned at Sotheby's on YouTube]
 - ▶ *Dutch auction* (= open descending-price auction):
price is lowered openly until some player stops the auction;
she receives the object and pays the stop price
[→ see Aalsmeer Flower Auction on YouTube]
 - ▶ *Sealed-bid first-price auction*:
all players simultaneously submit a bid to the auctioneer;
highest bidder receives the object and pays her bid
 - ▶ *Vickrey auction* (= sealed-bid second-price auction):
as in the first-price auction, but winner pays the *second-highest* bid
- Ties typically are broken uniformly or based on timing of bids

Revenues (in euros per capita) from 3G Auctions

TABLE 5.1
Revenues (in euros per capita) from European 3G Mobile Spectrum Auctions

<i>Year 2000</i>		<i>Year 2001</i>	
Austria	100	Belgium	45
Germany	615	Denmark	95
Italy	240	Greece	45
Netherlands	170		
Switzerland	20		
United Kingdom	650		

Source: Klemperer, P. (2004). Auctions: Theory and Practice. (p. 152)

Look up your valuation

Value for item auctioned off: look at the value next to your number

Number	Value	Number	Value	Number	Value	Number	Value
1	34.48	26	66.54	51	55.26	76	6.13
2	90.53	27	63.47	52	37.12	77	14.49
3	92.18	28	68.62	53	91.75	78	84.74
4	51.67	29	99.68	54	57.13	79	24.13
5	27.57	30	99.51	55	25.09	80	8.84
6	34.34	31	73.76	56	26.79	81	99.11
7	86.35	32	28.91	57	37.00	82	59.30
8	46.66	33	42.80	58	58.09	83	80.19
9	2.17	34	67.35	59	15.97	84	1.74
10	92.16	35	99.54	60	7.32	85	27.39
11	19.03	36	91.48	61	18.82	86	73.98
12	78.18	37	7.08	62	4.97	87	79.40
13	67.94	38	56.23	63	17.77	88	81.77
14	44.88	39	14.74	64	93.59	89	12.32
15	1.86	40	13.87	65	79.38	90	9.56
16	61.11	41	64.42	66	28.22	91	18.05
17	14.57	42	67.91	67	62.38	92	31.52
18	41.09	43	15.06	68	93.97	93	33.84
19	39.51	44	45.85	69	34.77	94	35.45
20	5.78	45	99.17	70	0.11	95	10.10
21	84.87	46	79.26	71	45.28	96	75.78
22	76.05	47	7.77	72	17.38	97	87.92
23	2.37	48	88.23	73	35.44	98	8.78
24	72.49	49	60.07	74	60.73	99	30.51
25	77.82	50	59.65	75	88.43	100	46.67

A short experiment - 1

QR code: <https://forms.gle/EoeAAwAtHk22v8SD6>



- We run a first-price auction
- You win if you have the highest bid (ties are randomly broken)
- Your payoff if you win: **your value - your bid**
- Values are uniformly distributed between 0 and 100
- Find your value:
 - ▶ Take the second and fourth digit of your birth date ddmm to form a number: xdxm → dm
 - ▶ Example: 10 April: 1004 → number: 04
 - ▶ Look up your value in the table above

A short experiment - 2

QR code: <https://forms.gle/rj7XxqLSEepw8wuT9>



- We run a second-price auction
- You win if you have the highest bid (ties are randomly broken)
- Your payoff if you win: **your value - the bid of the second highest bidder**
- Values are uniformly distributed between 0 and 100
- Find your value:
 - ▶ Take the second and fourth digit of your birth date ddmm to form a number: $x d x m \rightarrow d m$
 - ▶ Example: 10 April: 1004 $\rightarrow$ number: 04
 - ▶ Look up your value in the table above

A basic auction model

- The *symmetric independent private values (SIPV)* auction model assumes:
 - ▶ A seller wants to sell a single unit of an **indivisible object**
 - ▶ There are N potential buyers (= **bidders**) who have no budget constraints
 - ▶ Bidder i 's respective **value** x_i for the object is a random variable, distributed on $[0, \omega_i]$
 - ▶ Bidders' valuations are private information
 - ▶ All bidders' values a priori are independent and identically distributed (= i.i.d.) with cumulative distribution function $F_i = F$ and density $f_i = f$ (thus, $\omega_i = \omega$)
 - ▶ A strategy for bidder i is a **bidding function** $\beta_i : [0, \omega_i] \rightarrow \mathbb{R}_+$.
 - ▶ The seller is committed to the auction rules and has no special preference for any particular bidder
 - ▶ We will additionally assume that all bidders and the seller are risk-neutral;
in particular, bidder i gets utility $x_i - p$ if she wins at price p , and 0 if someone else wins

Second price sealed bid auction

- Payoff of a bidder:

$$\Pi_i = \begin{cases} v_i - \max_{j \neq i} b_j & \text{if } b_i > \max_{j \neq i} b_j \\ 0 & \text{if } b_i < \max_{j \neq i} b_j \\ \frac{v_i - b_i}{\# \text{highest bidders}} & \text{if } b_i = \max_{j \neq i} b_j \end{cases}$$

- **Result:** In a second-price sealed-bid auction, it is a weakly dominant strategy to bid according to $b(v_i) = v_i$

Second price sealed bid auction – analysis

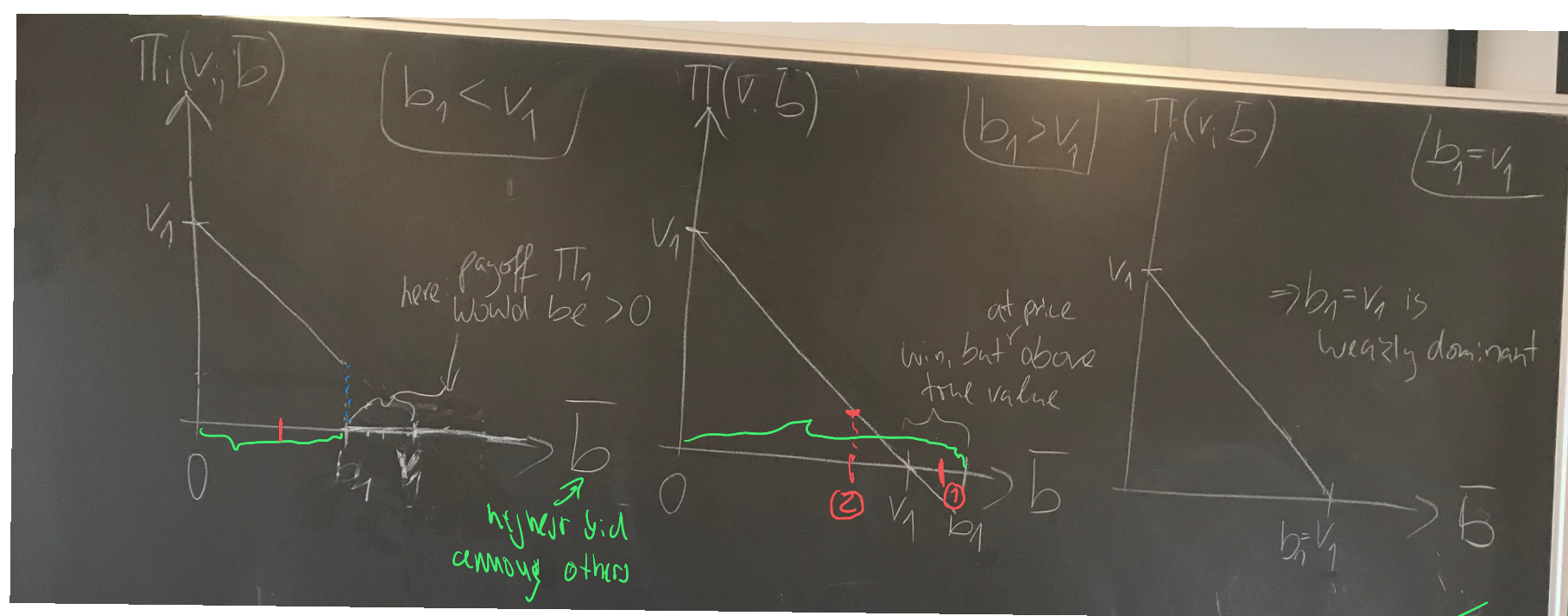
Let's show the above result:

- Consider bidder 1 with valuation v_1 and suppose $\bar{b} = \max_{j \neq 1} b_j$ is the highest competing bid
- We divide player 1's strategy space into three ranges:
 - ▶ submit bid $< v_1$
 - ▶ submit bid $> v_1$
 - ▶ submit bid $= v_1$
- What if he bids $b_1 < v_1$?
 - ▶ If $v_1 > b_1 \geq \bar{b}$ he still wins and his profit is still $v_1 - p_1$
 - ▶ If $\bar{b} > v_1 > b_1$ he still loses
 - ▶ If $v_1 > \bar{b} > b_1$ then he loses, whereas if he had bid v_1 he would have made a positive profit

Second price sealed bid auction – analysis

- What if he bids $b_1 > v_1$
 - Case ① in diagram on $b_1 > v_1$
 - ▶ If $b_1 \geq \bar{b} > v_1$: he ~~still~~ wins ^{but makes a loss!} and his profit is still $v_1 - p_1$
 - ▶ If $\bar{b} > b_1 > v_1$: he still loses
 - Case ② in diagram on $b_1 > v_1$
 - ▶ If $b_1 > v_1 > \bar{b}$: then he ^{still} wins, ~~but makes a loss, whereas if he had bid v_1 he would have made a positive profit~~
- What if he bids v_1 ?
 - ▶ By bidding v_1 bidder 1 will win if $v_1 > p_1$
 - ▶ He will lose if $v_1 < p_1$
 - ▶ If $v_1 = p_1$ he is indifferent between winning and losing

Second price sealed bid auction – graphical analysis



Vickrey format gives incentives for truth-telling $\rightarrow$ reveal their type ✓

Second price sealed bid auction – revenue

- Assume that each bidder's valuation is a uniformly distributed random variable: $V_i \sim U[0, 1]$
- What is the expected payment of a bidder if there are two bidders?

$$\text{Expected payment}_i(v_i) = \int_0^{v_i} f(v_j) v_j dv_j = \int_0^{v_i} v_j dv_j = \left[\frac{1}{2} v_j^2 \right]_0^{v_i} = \frac{1}{2} v_i^2$$

2nd equality: use that the values are drawn from the uniform distribution. Note that v_i is known to bidder i when he submits his bid.

- What is the expected revenue of the seller?

$$\begin{aligned} \text{Expected revenue} &= E [\text{Expected payment}_i(v_i) + \text{Expected payment}_j(v_j)] \\ &= E \left[\frac{1}{2} v_1^2 + \frac{1}{2} v_2^2 \right] = \frac{1}{2} [E(v_1^2) + E(v_2^2)] = \frac{1}{2} \left[\int_0^1 f(v_1) v_1^2 dv_1 \right. \\ &\quad \left. + \int_0^1 f(v_2) v_2^2 dv_2 \right] = \frac{1}{2} \left[\left[\frac{1}{3} v_1^3 \right]_0^1 + \left[\frac{1}{3} v_2^3 \right]_0^1 \right] = \frac{1}{2} \left[\frac{1}{3} + \frac{1}{3} \right] = \frac{1}{3} \end{aligned}$$

3rd equality: use that the valuations are independently drawn; 5th equality: use that the values are drawn from a uniform distribution

First price sealed bid auction

- Suppose there are 2 bidders participating in a first-price auction
- **Payoff** of a bidder:

$$\Pi_i = \begin{cases} v_i - b_i & \text{if } b_i > \max_{j \neq i} b_j \\ 0 & \text{if } b_i < \max_{j \neq i} b_j \\ \frac{v_i - b_i}{\# \text{highest bidders}} & \text{if } b_i = \max_{j \neq i} b_j \end{cases}$$

- A natural candidate for a **bidding function** (strategy): $b(v_i) = \alpha_i v_i$
- Suppose there exists a symmetric **Bayesian Nash equilibrium** in linear bidding functions. What does it look like?

First price sealed bid auction – analysis

- Bidder i 's **expected payoff**

$$\begin{aligned}\bar{u}_i(b_i, b(v_j) = \alpha_j v_j, v_i) &= \text{Prob}(\text{win}) \times \text{payoff} \\ &= \text{Prob}(b_i > b(v_j)) (v_i - b_i) = \text{Prob}(b_i > \alpha_j v_j) (v_i - b_i) \\ &= \text{Prob}\left(v_j < \frac{b_i}{\alpha_j}\right) (v_i - b_i) = F\left(\frac{b_i}{\alpha_j}\right) (v_i - b_i) \\ &= \frac{b_i}{\alpha_j} (v_i - b_i)\end{aligned}$$

$v_i, v_j \sim U[0,1]$

- Note that $\frac{b_i}{\alpha_j} \in [0, 1]$, so interpretation as the probability of winning is OK:
 - ▶ It is foolish for player i to bid above j 's maximum bid (α_j).
 - ▶ And pointless to bid below bidder j 's minimum bid (0)
 - ▶ Hence, $0 \leq b_i \leq \alpha_j$ and $0 \leq \frac{b_i}{\alpha_j} \leq 1$

First price sealed bid auction – analysis

- Expected payoff:

$$\max_{b_i \in [0, \alpha_j]} \bar{u}_i(b_i, b(v_j) = \alpha_j v_j, v_i) = \max_{b_i \in [0, \alpha_j]} \frac{b_i}{\alpha_j} (v_i - b_i)$$

- FOC:

$$\frac{1}{\alpha_j} (v_i - 2b_i) = 0 \quad \Leftrightarrow \quad b_i^* = \frac{1}{2} v_i,$$

i.e., $\alpha_j = \frac{1}{2}$

- Interpretation?

$\uparrow$
eq. strategy $b_i^*(v_i)$

General case:

$$b_i(v_i) = \frac{N-1}{N} v_i$$

more rivals $\rightarrow$ closer to true valuation

Expected revenue

- Exercise (for the first price sealed bid auction):
 - ▶ What is the expected payment of a bidder if there are two bidders?
 - ▶ What are the expected revenues of the seller?
- Expected revenues for the seller/payments by a bidder are equal in the two auction formats
- This is a more general principle ("*Revenue Equivalence Theorem*", see Tadelis ch. 13.1.4)]

Two equivalences

≈ "Japanese" auction

- In an **English auction** dropping out when the price reaches one's value is weakly dominant strategy, i.e., an English auction is strategically equivalent to a second-price sealed-bid auction
 - In a **Dutch auction** raising one's hand when the price reaches $\frac{1}{2} v_i$ whenever one's value is v_i is a symmetric BNE, i.e., a Dutch auction is strategically equivalent to a first-price sealed-bid
- In the SIPV model, it does not matter which of the four auction formats the seller uses